

A Novel Beamforming Approach for Bit-Level Phase Shifter Failures

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Abstract—In this paper, we address the problem of beamforming degradation in phased arrays using low-resolution digital phase shifters subject to bit-level hardware faults. Unlike prior work that models faults as complete element loss or random phase deviations on the element-level, we consider a scenario where specific bits of a phase shifter become stuck due to PIN diode malfunctions or switched-line trace damages. This fault mode results in a constrained, element-specific set of achievable discrete output phases. We propose a simple fault-aware beamforming algorithm called Element-wise Local Optimization (ELO) that adjusts the remaining functional bits of each faulty phase shifter to approximate the ideal phase as closely as possible. The algorithm operates in real-time under known fault conditions and does not require hardware replacements. Simulation results on a 1x8 uniform linear array demonstrate that the proposed method effectively restores the ideal radiation pattern, offering a lightweight solution for robust beamforming in cost-constrained and fault-aware systems. Comparison against literature baselines such as the Genetic Algorithm (GA) and Binary Particle Swarm Optimization (BPSO) show that the simple algorithm achieves comparable recovery with drastically shorter execution times.

Index Terms—Phased arrays, beamforming, optimization, radiation pattern recovery, low-resolution phase shifters

I. INTRODUCTION

PHASED array antennas are critical components in modern communication and sensing systems, offering dynamic beam steering and spatial filtering through controlled phase shifts at individual antenna elements. In cost and power-constrained environments where low-resolution phase shifters, often implemented using PIN diode-based switched networks, system robustness becomes increasingly sensitive to component-level degradation [1], [2], [3].

Recent work has focused on improving the robustness of phased arrays under failure conditions. Munsif and Ullah modeled failures as deviations of entire phase shifters to random phase states and employed multi-objective particle swarm optimization (MOPSO) to compensate using the remaining functional elements [4]. Tang *et al.* considered phase deviations introduced by antenna surface deformations [5]. Zhou *et al.* considered radiation pattern impairment due to structural degradation, such as radome ablation [6]. Other studies have considered total or random failure modes and applied global optimization techniques—such as particle swarm optimization (PSO), genetic algorithms (GA), and iterative Fourier techniques (IET)—to reconfigure the array [7], [8], [9]. These

optimization efforts were performed over the remaining phase shifters and have proven to be effective given enough time for convergence. However, they do not consider constrained bit-level behavior specific to digital phase shifters.

In contrast to existing models that address faults occurring at the element-level, phase-level or pattern-level, this paper introduces a new fault paradigm specific to low-resolution phase shifters: partial digital failures in which individual bits become stuck due to fragile diode faults. Such partial faults result in a constrained set of achievable output phases per element, depending on which bits remain operable. This nuanced failure mode, though realistic in hardware implementations, has not been addressed in prior literature [10].

We propose a simple fault-aware beamforming algorithm that operates offline, prior to deployment, once the positions and failure modes of the affected phase shifters are known. Rather than globally optimizing over the entire array, we locally adjust the remaining functional bits of the affected phase shifters to produce an output phase as close as possible to the desired ideal phase. By preserving the working hardware and adapting only the control logic, this approach offers a lightweight and hardware-efficient solution for restoring the beam pattern without requiring component replacement or real-time recalibration.

The proposed method is validated through numerical simulations that demonstrate its ability to recover beamforming performance under a variety of partial fault configurations. This was compared against widely-used global optimizations in the literature, and shown to achieve similar performance within a fraction of the time. This contribution opens new pathways for robust beamforming in constrained, low-resolution systems with known pre-mission hardware defects.

II. SYSTEM MODEL

A. Beamforming with Quantized Phase Shifters

We consider a uniform linear array (ULA) consisting of N isotropic antenna elements arranged along the x -axis with half-wavelength spacing $d = \lambda/2$. Each element is equipped with an M -bit digital phase shifter, allowing for 2^M discrete phase states uniformly distributed over the interval $[0, 2\pi)$. The ideal phase shift resolution is thus $\Delta\phi = \frac{2\pi}{2^M} = \frac{\pi}{2^{M-1}}$.

Let θ denote the angle of measurement from broadside, and let the desired beam steering direction be θ_0 . We assume the ideal progressive phase shift across elements is given by:

$$\phi_{\text{ideal}}^{(n)} = -kdn \sin \theta_0 = -\pi n \sin \theta_0, \quad n = 0, 1, \dots, N-1$$

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where $k = \frac{2\pi}{\lambda}$ is the wavenumber. The ideal (normalized) array factor can be expressed as:

$$AF_{\text{ideal}}(\theta) = \frac{1}{N} \sum_{n=0}^{N-1} w_n e^{jn\pi \sin \theta} \quad (1)$$

where $w_n = e^{j\phi_{\text{ideal}}^{(n)}}$ is the complex excitation weight, assumed to be of unit magnitude for all elements.

Each ideal phase $\phi_{\text{ideal}}^{(n)}$ must be quantized to the nearest representable value given the M -bit resolution. The set of quantized phase values is:

$$\mathcal{Q} = \left\{ 0, \frac{\pi}{2^{M-1}}, \frac{\pi}{2^{M-2}}, \frac{3\pi}{2^{M-1}}, \dots, \frac{(2^M - 1)\pi}{2^{M-1}} \right\}$$

We represent each M -bit phase setting as a binary word $\mathbf{b} = b_{M-1} \dots b_2 b_1 b_0$ that would reproduce the quantized phase, with b_{M-1} representing the most significant bit (MSB). The actual quantized phase applied to the n -th element is:

$$\phi^{(n)} = \frac{\pi}{2^{M-1}} \cdot \sum_{i=0}^{M-1} b_i^{(n)} \cdot 2^i \quad (2)$$

B. Fault Model

We consider two types of hardware faults in the digital phase shifters:

- **MSB Failures:** The most significant bit of certain elements (i.e., b_{M-1}) are stuck at either 0 or 1 due to diode malfunction. This makes an entire half-cycle of potential phase shifts inaccessible to the phase shifter. This failure mode becomes increasingly likely for low-resolution phase shifters (e.g., $M = 4$).
- **Random Bit Failures:** A more general fault model in which any subset of bits (not necessarily contiguous or most significant) may be stuck, and the failure pattern is randomly assigned across the array.

In general, for each element n there is a set of bit positions $\mathcal{P}_0^{(n)}$ with values stuck at 0, and a set of bit positions $\mathcal{P}_1^{(n)}$ with values stuck at 1. This overrides the element phase to:

$$\phi_{\text{faulty}}^{(n)} = \frac{\pi}{2^{M-1}} \cdot \left(\sum_{\substack{i=0 \\ i \notin \mathcal{P}_0^{(n)} \cup \mathcal{P}_1^{(n)}}}^{M-1} b_i^{(n)} \cdot 2^i + \sum_{i \in \mathcal{P}_1^{(n)}} 2^i \right) \quad (3)$$

which distorts the array factor and hence radiation pattern.

We assume that phase settings had been optimized for the array prior to the bit-level faults, using any of the optimization algorithms in existing literature, and stored in a code-book. The goal of the proposed self-repair algorithm is to minimize the distortion in the radiation pattern away from this optimized pattern, due to the bit-level faults. This algorithm is run offline assuming the failure state of each phase shifter is known prior to deployment during hardware inspection.

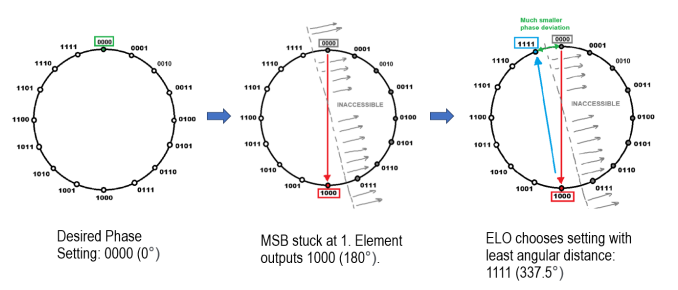


Fig. 1. An example 4-bit phase shifter state. Suppose the desired phase setting is 0000 and the MSB is stuck at 1, resulting in an output phase of 1000 - a half cycle phase error. ELO discovers the phase setting closest to 0000 in wrapped angular distance that also satisfies the faulty bit constraint - 1111, and overwrites the phase setting.

III. PROPOSED ALGORITHM

Given the fault state, each element n has a constrained set of achievable phase values $\mathcal{Q}_{\text{avail}}^{(n)} \subseteq \mathcal{Q}$ and a set of binary words $\mathcal{B}_{\text{avail}}^{(n)}$ that produce those phase values.

Given the existing optimization codebook during operation, each element has a desired quantized phase value $\phi_{\text{desired}}^{(n)}$ produced by the binary word $\mathbf{b}_{\text{desired}}^{(n)}$.

Suppose $\phi_{\text{desired}}^{(n)} \notin \mathcal{Q}_{\text{avail}}^{(n)}$. In Algorithm 1, the reachable phase ϕ^* with the minimum angular distance modulo 2π to the desired phase is chosen to replace the desired phase. This corresponds to the $\mathbf{b}^* \in \mathcal{B}_{\text{avail}}^{(n)}$ with the minimum distance modulo 2^M to $\mathbf{b}_{\text{desired}}^{(n)}$.

The resulting array factor becomes:

$$AF_{\text{repaired}}(\theta) = \frac{1}{N} \sum_{n=0}^{N-1} e^{j(\phi^{*(n)} + kdn \sin \theta)}$$

The simple working principle is illustrated with an example phase shifter in Figure 1

Algorithm 1 Element-wise Local Optimization (ELO)

Input: Desired beamforming vector $\mathbf{w} \in C^N$, phase shifter resolution M , sets of bit positions stuck at 0 for each element $\{\mathcal{P}_0^{(1)}, \mathcal{P}_0^{(2)} \dots \mathcal{P}_0^{(N)}\}$, sets of bit positions stuck at 1 for each element $\{\mathcal{P}_1^{(1)}, \mathcal{P}_1^{(2)} \dots \mathcal{P}_1^{(N)}\}$

Output: Repaired beamforming vector $\mathbf{w}^* \in C^N$

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 $\mathbf{w}^* \leftarrow \mathbf{w}$ 
 $\Delta\phi \leftarrow 2\pi/2^M$ 
for  $n = 1$  to  $N$  do
   $\phi_{\text{desired}} \leftarrow \angle w_n$ 
   $\mathbf{b}_{\text{desired}} \leftarrow \text{DecToBin}(\text{Round}(\phi_{\text{desired}}/\Delta\phi))$ 
   $\mathcal{B} \leftarrow \text{AvailableBinaryWordSet}(M, \mathcal{P}_0^{(n)}, \mathcal{P}_1^{(n)})$ 
  if  $\mathbf{b}_{\text{desired}} \notin \mathcal{B}$  then
     $\mathbf{b}^* \leftarrow \arg \min_{\mathbf{b} \in \mathcal{B}} \text{WrappedDistance}(\mathbf{b}, \mathbf{b}_{\text{desired}})$ 
     $\phi^* \leftarrow \Delta\phi \times \text{BinToDec}(\mathbf{b}^*)$ 
     $w_n^* \leftarrow e^{j\phi^*}$ 
  end if
end for
return  $\mathbf{w}^*$ 

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IV. SIMULATION AND RESULTS

A 1×8 half-wavelength ($d = \lambda/2$) ULA with $M = 4$ -bit phase shifters ($\Delta\phi = 22.5^\circ$) was steered to $\theta_0 \in [-90^\circ, 90^\circ]$. To isolate the impact of bit-level faults, we disabled mutual coupling, amplitude tapering and noise.

For a given fault cardinality F (number of stuck bits in the entire array), we generated 1000 Monte-Carlo trials:

- 1) **Fault assignment.** F bit positions were selected uniformly at random across the 8 phase shifters, and assigned either 0 or 1. The MSB failures and random bit failures were considered separately.
- 2) **Beamformer synthesis.** The ideal progressive phases for the steering angle were quantized to the nearest 4-bit value, providing the reference beamforming vector $\mathbf{w}_{\text{ideal}}$ and the resultant array factor AF_{ideal} .
- 3) **Fault injection.** Each selected faulty bit was overwritten with their assigned value, yielding the faulty vector $\mathbf{w}_{\text{faulty}}$ and the array factor AF_{faulty} .
- 4) **Repair and compare.** ELO as described in Algorithm 1 was applied to obtain the repaired vector $\mathbf{w}^*$ and the array factor AF^* . The Genetic Algorithm (GA) and the Binary Particle Swarm Optimization (BPSO) [11] were also applied to characterize a baseline of the literature for ELO to compare against. BPSO was chosen instead of PSO due to the binary search space associated with digital phase shifter arrays.

The array factors AF_{faulty} and AF^* were evaluated on a 1° grid, and for each of them three scalar metrics were extracted:

$$\text{PBP} = 20 \log_{10} |AF(\theta_0)| \quad (4)$$

$$\text{MSE}_{\text{norm}} = \frac{\sum_i |AF(\theta_i) - AF_{\text{ideal}}(\theta_i)|^2}{2 \sum_i |AF_{\text{ideal}}(\theta_i)|^2} \quad (5)$$

$$\text{PSLL} = \frac{20 \log_{10} |\max_{\theta} AF(\theta)|}{\text{PBP}} \quad (6)$$

Peak beam power (PBP) captures main-beam preservation, MSE_{norm} measures the global normalized energy displacement, and Peak Side-Lobe Level (PSLL) describes the level of energy coming from the most significant unwanted direction. All numerical simulation was carried out in Python.

A. Impact of Bit Significance

Figure 2 compares typical radiation patterns obtained with *MSB-only* and *random* faults. Although both cases involve the same number of faulty bits, the MSB fault severely distorts the main lobe and raises sidelobes relative to the random fault. The observation is repeated across the Monte-Carlo ensemble.

B. Radiation Pattern Recovery Effectiveness

Figure 3 illustrates the repaired pattern produced by ELO for a typical fault realization. The algorithm almost fully restores the original pattern.

The complete performance profile of the three algorithms under all fault cardinalities for the 1×8 ULA is shown in Figure 4, where the domain is the proportion of faulty bits

$$r := F/(MN).$$

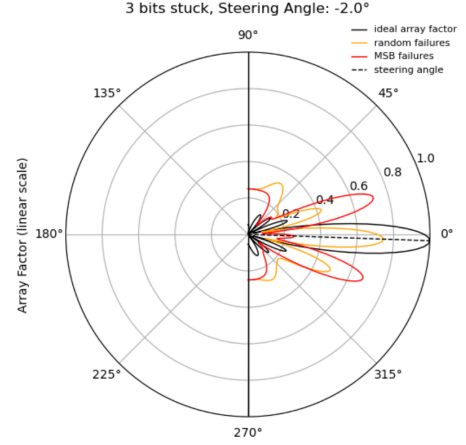


Fig. 2. A typical realization of 3 stuck phase shifter bits, in random failure mode and in MSB mode.

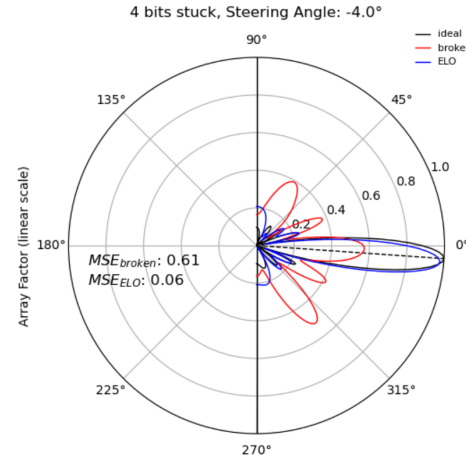


Fig. 3. Radiation pattern before and after element-wise repair for a representative random fault realization $F = 4$.

The domain consists of 3 regions:

- $r < 0.2$: The region of practical interest. Here, GA resulted in the lowest MSE_{norm} and PSLL, and the highest PBP. ELO was suboptimal, but still significantly reduced MSE_{norm} and PSLL, and significantly restored PBP. BPSO was the least performant for most of this region, but overtakes ELO in all metrics at the upper end of the region.
- $0.2 \leq r < 0.6$: ELO becomes the least effective algorithm in this region while the BPSO approaches the performance of GA.
- $r > 0.6$: BPSO has converged with GA, while ELO remains the least effective. None of the algorithms can recover the radiation pattern to a usable state.

C. Execution Time

The optimization time for each algorithm was averaged in all tests and is presented in Table I. ELO was significantly faster than GA and BPSO.

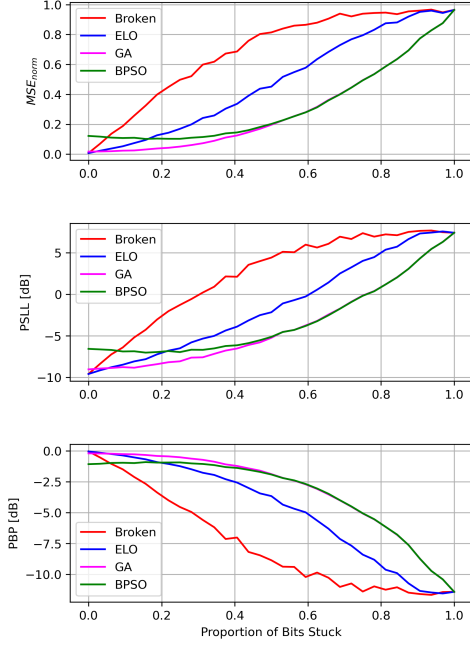


Fig. 4. Aggregated results of the Monte Carlo trials. The domain of the plot is the proportion of bits stuck in the entire array, obtained by $F/(MN)$.

TABLE I
AVERAGE EXECUTION TIME OF EACH ALGORITHM

ELO	GA	BPSO
0.2052	4171.7777	4601.8420

D. Discussion

Bit-level faults were found to severely damage the radiation pattern even for a low number of faulty bits. MSB failure mode had more severe damages compared to the random failure mode.

The proposed ELO algorithm, being very simple, was able to quickly recover most of the radiation pattern for practical fault cardinalities. This performance closely matches with GA, and even outperforms BPSO. The low runtime allows the ELO to run in real time, should fault-monitoring systems detect bit-level faults during deployment.

With these results, the aim of developing a fault-tolerant smart beamforming approach to address bit-level faults in the phase shifters was achieved, and validated with simulation.

E. Limitations and Future Work

The study assumed uniform random stuck-bit distributions due to a lack of empirical failure rate data for PIN diode-based phase shifters. In practice, some bits may be more prone to failure due to thermal stress or layout-specific vulnerabilities. The recovery algorithm was tested against a generic fault case, which may not align perfectly with real-world failure distributions. This assumption does not invalidate the results but merely leads to a lack of precise expectation for real world fault levels. Since the study was conducted over all fault states, this was not a significant problem. Conducting accelerated

aging experiments could provide more realistic failure models, enabling the development of more targeted recovery strategies.

V. CONCLUSION

This work investigated the impact of bit-level faults in digital phase shifters on phased array antennas and introduced the Element-wise Local Optimization (ELO) algorithm to mitigate these effects in real time. By explicitly modeling partial bit failures of phase shifters and exploiting remaining bit-level degrees of freedom, ELO offers a simple yet effective strategy for fault-aware beamforming, addressing a gap in existing literature that largely overlooks bit-level failure modes. Its performance validated against global optimizers like GA and BPSO. ELO demonstrated competitive or superior results with significantly lower runtimes due to its simplicity, enabling dynamic in-field adaptation to faults.

This contribution enhances the theoretical understanding of digital phase shifter faults and presents a practical, deployable method to improve system resilience without hardware changes. Future work could focus on empirical fault data collection and exploring real-time diagnostics and adaptive antenna design.

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